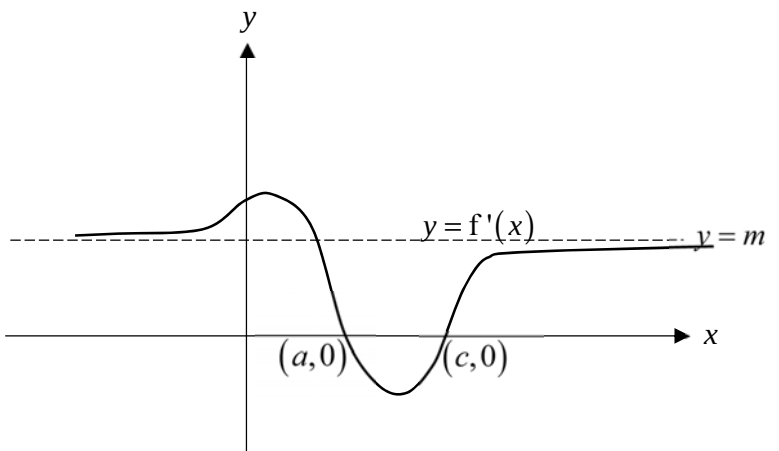
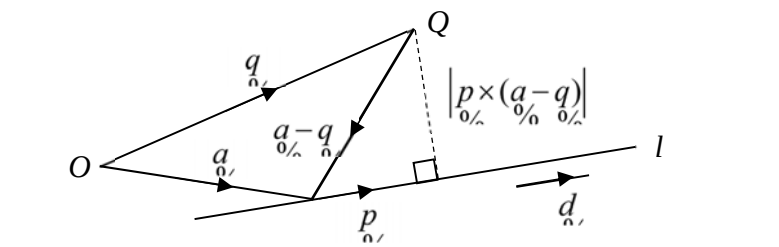
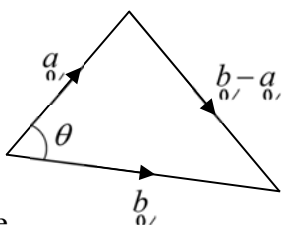


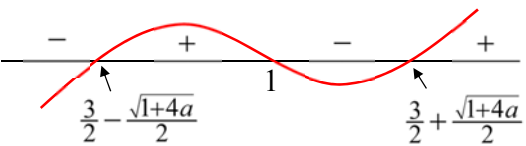
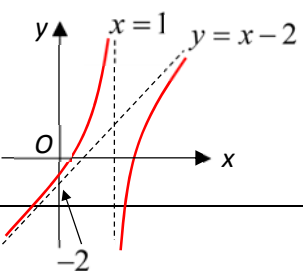
Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
1 [2]		G1 – Shape with at least 2 features correct G1 – all features correct SR: The maximum point could be in the first or second quadrant 1. x-intercept $(a, 0)$ 2. x-intercept $(c, 0)$ 3. H.A $y = m$ (above x-axis) 4. y-intercept above H.A $y = m$
2 [4]	Let required equation be $y = ax^3 + bx^2 + cx + d$, $a, b, c, d \in \mathbb{R}$. Substitute $(0, 0)$ into equation, $\therefore d = 0$ $0.15^3 a + 0.15^2 b + 0.15c = 0.1$ $0.3^3 a + 0.3^2 b + 0.3c = 0.125$ $0.6^3 a + 0.6^2 b + 0.6c = 0.1$ Using GC, $a = \frac{50}{27}$, $b = -\frac{5}{2}$, $c = 1$ Required equation is $y = \frac{50}{27}x^3 - \frac{5}{2}x^2 + x \dots (*)$ Substitute $x = 0.45$ into $(*)$, $y = 0.1125$ $\therefore$ the manufacturer's claim is accurate.	B1 – $d = 0$ [SOI] M1 – Formulate system of equations with at least 1 equation correct A1 – Correct equation B1 – Correct conclusion with substitution of $x = 0.45$ into $(*)$ (SOI)
3(a) [1]	 $\left \frac{p \times (a - q)}{ a } \right$ is the shortest distance (or perpendicular distance) from Q to l.	B1 – Correct Geometrical Interpretation

Marking Scheme for HCI 2018 Prelim Paper 1

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	OR $\left \begin{matrix} p \times (a-q) \end{matrix} \right$ is the area of the parallelogram with adjacent vectors $\begin{matrix} p \\ \end{matrix}$ and $\begin{matrix} a-q \\ \end{matrix}$.	
3(b) [4]	 By cosine rule, $\begin{aligned} b-a ^2 &= a ^2 + b ^2 - 2 a b \cos\theta \\ (b-a) \cdot (b-a) &= a \cdot a + b \cdot b - 2 a b \cos\theta \\ b \cdot b - b \cdot a - a \cdot b + a \cdot a &= a \cdot a + b \cdot b - 2 a b \cos\theta \\ -2a \cdot b &= -2 a b \cos\theta \\ a \cdot b &= a b \cos\theta \quad (\text{shown}) \end{aligned}$	B1 – Writing 3rd side of Δ as $\begin{matrix} b-a \\ \end{matrix}$ M1 – Apply cosine rule to obtain $\begin{matrix} b-a ^2 = a ^2 + b ^2 - 2 a b \cos\theta \end{matrix}$ M1 – Write $\begin{matrix} b-a ^2 \end{matrix}$ as $\begin{matrix} (b-a) \cdot (b-a) \end{matrix}$ and expand AG1 – Obtain $\begin{matrix} -2a \cdot b = -2 a b \cos\theta \end{matrix}$ and show that $\begin{matrix} a \cdot b = a b \cos\theta \end{matrix}$
4(i) [7]	$\sqrt{\frac{1}{4}} = \frac{1}{2}$ Since $a > 0$, $a + \frac{1}{4} > \frac{1}{4} > 0$ $\sqrt{a + \frac{1}{4}} > \frac{1}{2}$ $-\sqrt{a + \frac{1}{4}} < -\frac{1}{2}$ $\therefore \frac{3}{2} - \sqrt{a + \frac{1}{4}} < 1 \quad (\text{shown})$ $a + \frac{1}{4} > \frac{1}{4} > 0$	AG1 – Make use of $a > 0$, compare $\sqrt{a + \frac{1}{4}} > \frac{1}{2}$ to show $\frac{3}{2} - \sqrt{a + \frac{1}{4}} < 1$
(ii)	$\frac{x^2 - x - a}{x-1} \leq 2$ $\frac{x^2 - x - a}{x-1} - 2 \leq 0$	M1 – Place all terms on one side and take common denominator (Algebraic method)

Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
	$\frac{x^2 - x - a - 2x + 2}{x-1} \leq 0$ $\frac{x^2 - 3x + 2 - a}{x-1} \leq 0$ <u>Method 1:</u> (Algebraic method) $\frac{(x - \frac{3}{2})^2 - (\frac{3}{2})^2 + 2 - a}{x-1} \leq 0$ $\frac{(x - \frac{3}{2})^2 - (\frac{1}{4} + a)}{x-1} \leq 0$ $\frac{(x - \frac{3}{2})^2 - (\sqrt{\frac{1+4a}{4}})^2}{x-1} \leq 0$ $\frac{(x - \frac{3}{2} + \frac{\sqrt{1+4a}}{2})(x - \frac{3}{2} - \frac{\sqrt{1+4a}}{2})}{x-1} \leq 0$ $\frac{\left[x - \left(\frac{3}{2} - \sqrt{\frac{1}{4} + a} \right) \right] \left[x - \left(\frac{3}{2} + \sqrt{\frac{1}{4} + a} \right) \right]}{x-1} \leq 0$ Since a is a positive constant, $\left(\frac{3}{2} - \sqrt{\frac{1}{4} + a} \right) < 1$ (from (i)) And $\left(\frac{3}{2} + \sqrt{\frac{1}{4} + a} \right) > \frac{3}{2} > 1$  $\therefore x \leq \frac{3}{2} - \frac{\sqrt{1+4a}}{2}$ or $1 < x \leq \frac{3}{2} + \frac{\sqrt{1+4a}}{2}$ <u>Method 2:</u> (Graphical method) $x - 2 - \frac{a}{x-1} \leq 0$  $\begin{array}{r} x-2 \\ x-1 \overline{) x^2 - 3x + 2 - a} \\ \underline{x^2 - x} \\ -2x + 2 - a \\ \underline{-2x + 2} \\ -a \end{array}$	M1 – Attempt to factorise $x^2 - 3x + 2 - a$ by completing the square or using formula M1 – Find all the critical values and attempt to using a number line determine range of x A1 – Correct answer (Graphical method) M1 – Attempt to sketch graph to locate parts of the graph below x-axis M1 – Attempt to find x-intercepts of graph A1 – Correct answer

Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
	Let $\frac{x^2 - 3x + 2 - a}{x - 1} = 0$ $\therefore x^2 - 3x + 2 - a = 0$ $\therefore x = \frac{3 \pm \sqrt{9 - 4(2 - a)}}{2}$ $= \frac{3}{2} \pm \frac{\sqrt{1 + 4a}}{2}$ Since $\frac{x^2 - 3x + 2 - a}{x - 1} \leq 0$, $\therefore x \leq \frac{3}{2} - \frac{\sqrt{1 + 4a}}{2}$ or $1 < x \leq \frac{3}{2} + \frac{\sqrt{1 + 4a}}{2}$	
	Replace x in $\frac{x^2 - x - a}{x - 1} \leq 2$ with $1 - x$, we obtain $\frac{(1 - x)^2 - (1 - x) - a}{(1 - x) - 1} \leq 2$ $\frac{1 - 2x + x^2 - 1 + x - a}{-x} \leq 2$ $\frac{x^2 - x - a}{-x} \leq 2$ $\frac{a + x - x^2}{x} \leq 2$ Hence $1 - x \leq \frac{3}{2} - \frac{\sqrt{1 + 4a}}{2}$ or $1 < 1 - x \leq \frac{3}{2} + \frac{\sqrt{1 + 4a}}{2}$ $\therefore x \geq -\frac{1}{2} + \frac{\sqrt{1 + 4a}}{2}$ or $-\frac{1}{2} - \frac{\sqrt{1 + 4a}}{2} \leq x < 0$	M1 – Replace x with $1 - x$ A1 – Correct answer
5(i) [8]	$f(r) - f(r - 1) = \cos[\alpha + (r + \frac{1}{2})\beta] - \cos[\alpha + (r - \frac{1}{2})\beta]$ $= -2\sin(\alpha + r\beta)\sin\frac{1}{2}\beta$ (By factor formula)	M1 – Apply factor formula A1 – Correct answer
5(ii)	$\sum_{r=0}^n [f(r) - f(r - 1)] = \sum_{r=0}^n [-2\sin(\alpha + r\beta)\sin\frac{1}{2}\beta]$ $= -2\sin\frac{1}{2}\beta \sum_{r=0}^n \sin(\alpha + r\beta)$	M1 – Able to use part (i)

Marking Scheme for HCI 2018 Prelim Paper 1

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	$\sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \dots + \sin(\alpha + n\beta)$ $= \sum_{r=0}^n \sin(\alpha + r\beta)$ $\sum_{r=0}^n \sin(\alpha + r\beta)$ $= -\frac{1}{2 \sin \frac{1}{2} \beta} \sum_{r=0}^n [f(r) - f(r-1)]$ $= -\frac{1}{2 \sin \frac{1}{2} \beta} \left[\begin{array}{l} f(0) - f(-1) \\ + f(1) - f(0) \\ + \dots \\ + f(n-1) - f(n-2) \\ + f(n) - f(n-1) \end{array} \right]$ $= -\frac{1}{2 \sin \frac{1}{2} \beta} [f(n) - f(-1)]$ $= -\frac{1}{2 \sin \frac{1}{2} \beta} \{ \cos[\alpha + (n + \frac{1}{2})\beta] - \cos(\alpha - \frac{1}{2}\beta) \}$ $= -\frac{1}{2 \sin \frac{1}{2} \beta} [-2 \sin(\alpha + \frac{n}{2}\beta) \sin(\frac{n+1}{2}\beta)]$ $= \frac{\sin(\alpha + \frac{n}{2}\beta) \sin(\frac{n+1}{2}\beta)}{\sin \frac{1}{2} \beta} \dots (1)$ Where $p = \frac{n}{2}$ and $q = \frac{n+1}{2}$.	M1 – Attempt to use MOD, must show the cancellation and at least 2 rows in front and at the end M1 – Apply factor formula A1 – Obtain a and b correctly
5(iii)	Sub $\beta = \pi$ Differentiate (1) wrt α : $\cos \alpha + \cos(\alpha + \pi) + \cos(\alpha + 2\pi) + \dots + \cos(\alpha + n\pi)$ $= \frac{\cos\left(\alpha + \frac{n\pi}{2}\right) \sin\left[\left(\frac{n+1}{2}\right)\pi\right]}{\sin \frac{\pi}{2}}$ $= \cos\left(\alpha + \frac{n\pi}{2}\right) \sin\left[\left(\frac{n+1}{2}\right)\pi\right]$	M1 – Sub $\beta = \pi$ and attempt to differentiate (1) wrt α A1 – Show all steps clearly with the correct answer

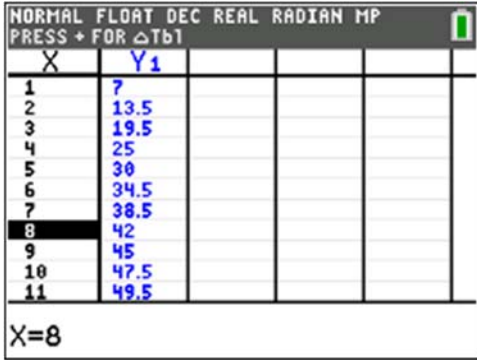
Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
	Alternative Solution: $\cos \alpha + \cos(\alpha + \pi) + \cos(\alpha + 2\pi) + \dots + \cos(\alpha + n\pi)$ $= \sum_{r=0}^n \cos(\alpha + r\pi)$ $= \sum_{r=0}^n \sin\left(\frac{\pi}{2} - \alpha - r\pi\right)$ $= \frac{\sin\left(\frac{\pi}{2} - \alpha - \frac{n\pi}{2}\right) \sin\left[-\pi\left(\frac{n+1}{2}\right)\right]}{\sin\left(-\frac{\pi}{2}\right)}$ $= \sin\left(\frac{\pi}{2} - \alpha - \frac{n\pi}{2}\right) \sin\left[\pi\left(\frac{n+1}{2}\right)\right]$ $= \cos\left(\alpha + \frac{n\pi}{2}\right) \sin\left[\pi\left(\frac{n+1}{2}\right)\right]$	M1 – Use of $\cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$, $\beta = -\pi$ A1 – Show all steps clearly with the correct answer
6(i) [4]	$\frac{dx}{dt} = 2 \cos t$ $\frac{dy}{dt} = -\sin t$ $\frac{dy}{dx} = \frac{-\sin t}{2 \cos t} = -\frac{\tan t}{2}$ Equation of tangent T is $y - 1 - \cos p = -\frac{\tan p}{2}(x - 2 \sin p)$ $y = 1 + \cos p - \frac{\tan p}{2}x + \frac{\sin^2 p}{\cos p}$ $= 1 + \frac{\cos^2 p + \sin^2 p}{\cos p} - \frac{\tan p}{2}x$ $= 1 + \sec p - \frac{\tan p}{2}x$ $2y + x \tan p = 2(1 + \sec p)$	M1 – Either $\frac{dx}{dt}$ or $\frac{dy}{dt}$ correct A1 – Correct $\frac{dy}{dx}$ in terms of t M1 – Form equation of tangent AG1 – Correct equation of T
6(ii) [5]	When $y = 0$, $\frac{\tan p}{2}x = 1 + \sec p$ $x = \frac{2 + 2 \sec p}{\tan p}$ When $x = 0$, $y = 1 + \sec p$	M1 – Substitute $x = 0$, $y = 0$ to find A and B. Also accept $x = 2 \cot p + 2 \operatorname{cosec} p$ or $x = \frac{2}{\tan p} + \frac{2}{\sin p}$.

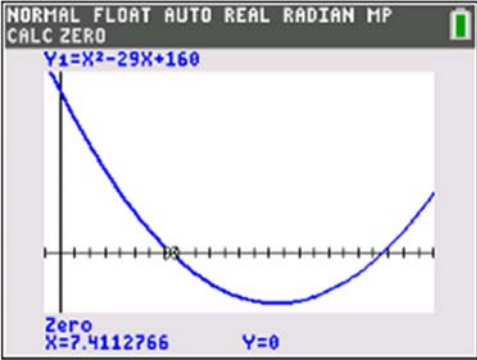
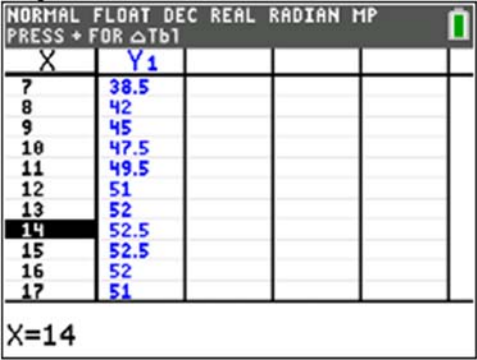
Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
	Method 1: Coordinates of $M = \left(\frac{1 + \sec p}{\tan p}, \frac{1 + \sec p}{2} \right)$. $y = \frac{1 + \sec p}{2} \Rightarrow \sec p = 2y - 1$ $x = \frac{1 + \sec p}{\tan p} = \frac{2y}{\tan p}$ Using $1 + \tan^2 p = \sec^2 p$, $1 + \left(\frac{2y}{x} \right)^2 = (2y - 1)^2$ $1 + \frac{4y^2}{x^2} = 4y^2 - 4y + 1$ $y = yx^2 - x^2$ $x^2 = y(x^2 - 1)$ $y = \frac{x^2}{x^2 - 1}$	M1 – Apply mid-point formula $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$ A1 – Obtain x and y coordinates of M. M1 – Form cartesian equation of locus of M using $1 + \tan^2 p = \sec^2 p$ A1 – Correct cartesian equation of locus of M.
	Method 2: Coordinates of $M = \left(\cot p + \operatorname{cosec} p, \frac{1 + \sec p}{2} \right)$. $y = \frac{1 + \sec p}{2} \Rightarrow \sec p = 2y - 1$ $x = \cot p + \operatorname{cosec} p = \frac{\cos p + 1}{\sin p}$ Using $\sin^2 p + \cos^2 p = 1$,	M1 – Express $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$ A1 – Obtain x and y coordinates of M. M1 – Form cartesian equation of locus of M using $\sin^2 p + \cos^2 p = 1$

Marking Scheme for HCI 2018 Prelim Paper 1

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	$\left(\frac{\cos p + 1}{x}\right)^2 + \left(\frac{1}{2y - 1}\right)^2 = 1$ $\frac{\left(\frac{1}{2y - 1} + 1\right)^2}{x^2} + \frac{1}{(2y - 1)^2} = 1$ $\frac{(2y)^2}{(2y - 1)^2 x^2} + \frac{1}{(2y - 1)^2} = 1$ $4y^2 + x^2 = (4y^2 - 4y + 1)x^2$ $y^2 = y^2 x^2 - yx^2$ $x^2 = y(x^2 - 1)$ $y = \frac{x^2}{x^2 - 1}$	A1 – Correct cartesian equation of locus of M .						
7 [11] (i)	By Plan A, area painted on the n th day $= 7 - (n - 1)0.5$ $= 7.5 - 0.5n$	B1 – Correct formula						
(ii)	Total area painted on n days $= \frac{n}{2}[2(7) - (n - 1)0.5] \geq 40$ Method 1:  <table border="1" data-bbox="197 1550 394 1662"><thead><tr><th>n</th><th>Area</th></tr></thead><tbody><tr><td>7</td><td>$38.5 < 40$</td></tr><tr><td>8</td><td>$42 > 40$</td></tr></tbody></table> He will finish painting his room on the 8th day.	n	Area	7	$38.5 < 40$	8	$42 > 40$	M1 – Form an inequality or use GC table involving the sum of AP formula SR: If students form equation instead of inequality, award 1 out of 2 marks. A1 – Correct answer
n	Area							
7	$38.5 < 40$							
8	$42 > 40$							
	Method 2: Total area painted on n days =	M1 – Form an inequality or use GC table involving the sum of AP formula						

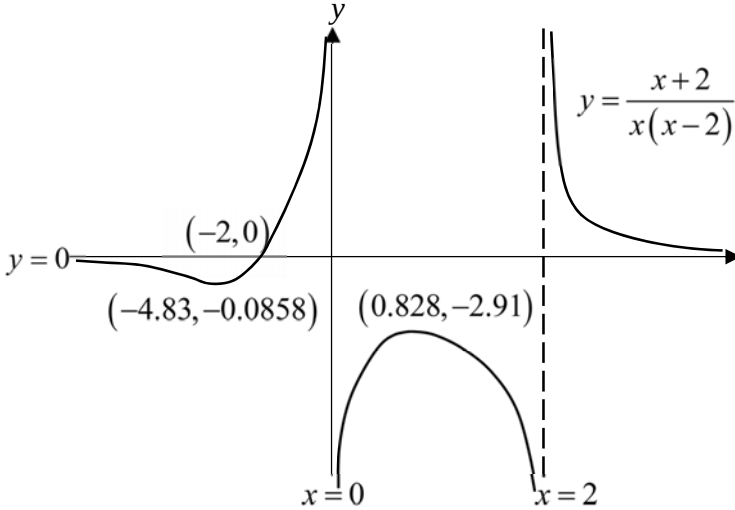
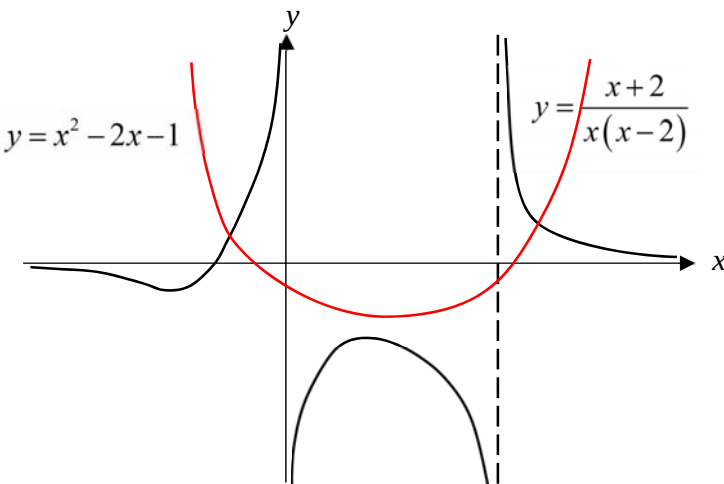
Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
	$\frac{n}{2}[2(7) - (n-1)0.5] \geq 40$ $28n - n(n-1) \geq 160$ $n^2 - 29n + 160 \leq 0$  $7.41 \leq n \leq 21.6$ But $7.5 - 0.5n > 0 \Rightarrow n < 15$ So $8 \leq n \leq 14$. Least $n = 8$. He will finish painting his room on the 8th day.	SR: If students form equation instead of inequality, award 1 out of 2 marks. A1 – Correct answer
(iii)	For $n \geq 15$, area painted on the nth day = 0 and painting stops. Therefore Plan A cannot be applied to an arbitrarily large wall.  X=14 By GC or using $\frac{14}{2}[2(7) - (14-1)0.5] = 52.5$, largest area that Plan A can be applied to is 52.5 m^2.	B1 – Correct reason B1 – Correct largest area.

Marking Scheme for HCI 2018 Prelim Paper 1

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(iv)	$7 \frac{1-0.8^n}{1-0.8} \geq 0.7 \times 40$ $\frac{1-0.8^n}{0.2} \geq 4$ $1-0.8^n \geq 0.8$ $0.8^n \leq 0.2$ $n \ln 0.8 \leq \ln 0.2$ $n \geq 7.21$ He needs 8 days to finish painting at least 70% of his room.	M1 – Apply correct GP formula for S_n M1 – Form inequality $S_n \geq 0.7 \times 40$ and attempt to solve algebraically. A1 – Correct number of days.
(v)	If he chooses Plan B, total area painted after an infinite number of days = $\frac{7}{1-0.8} = 35 < 40$. He cannot finish painting his room. Therefore he should choose Plan A.	B1 – Correct conclusion with reason with correct formula for sum to infinity used and the answer is < 40
8[10] (i)	$y = \frac{x+2}{x(x+k)}$ $\frac{dy}{dx} = \frac{x(x+k) - (x+2)(2x+k)}{[x(x+k)]^2}$ $= \frac{x^2 + kx - 2x^2 - kx - 4x - 2k}{[x(x+k)]^2}$ $= -\frac{x^2 + 4x + 2k}{[x(x+k)]^2}$ Let $\frac{dy}{dx} = 0 \Rightarrow x^2 + 4x + 2k = 0$ For no stationary points, Discriminant < 0 $16 - 4(2k)(1) < 0$ $k > 2$ When $k = 2$, $y = \frac{x+2}{x(x+2)} = \frac{1}{x}$, $x \neq -2$ $\Rightarrow$ No stationary points when $k = 2$ $\therefore k \geq 2$	M1 – Attempt to find $\frac{dy}{dx}$ using quotient/product rule M1 – Use Discriminant < 0 to obtain a linear inequality involving k A1 – correct answer $k \geq 2$

Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
8(ii)		G1 – Shape (with 2 turning points) G1 – at least 4 out of 6 features G1 – All correct 1. x-intercept $(-2, 0)$ 2. min $(-4.83, -0.0858)$ 3. max $(0.828, -2.91)$ 4. V.A $x = 2$ 5. V.A $x = 0$ (SOI) 6. H.A $y = 0$ (SOI)
8(iii)	$m = 0, -0.0858$ or -2.91 (to 3s.f)	B1 – All 3 values correct
8(iv)	$x^4 - 4x^3 + 3x^2 + x - 2 = 0$ $x^4 - 4x^3 + 3x^2 + 2x = x + 2$ $x(x-2)(x^2 - 2x - 1) = x + 2$ $x^2 - 2x - 1 = \frac{x+2}{x(x-2)}$ Sketch the graph of $y = x^2 - 2x - 1$ in (ii) 	M1 – attempt to find the expression of the second curve by rearranging the terms And sketch the correct curve $y = x^2 - 2x - 1$ AG1 – conclude with reference to diagram

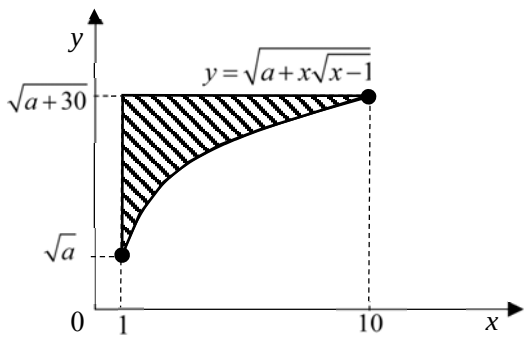
Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
	Since the graph $y = x^2 - 2x - 1$ intersect the graph $y = \frac{x+2}{x(x-2)}$ at two distinct points, there are 2 real roots to the equation $x^4 - 4x^3 + 3x^2 + x - 2 = 0$.	
9[12] (a)	$4iz - w = 9i - 13$ ----- (1) $(4 + 2i)w^* = z + 3i$ -----(2) Let $w = x + iy$ $4i[(4+2i)(x - iy) - 3i] - (x + iy) = 9i - 13$ $4i[(4x+2xi - 4iy + 2y - 3i) - x - iy] = 9i - 13$ $16ix - 8x + 16y + 8yi + 12 - x - iy = 9i - 13$ Compare real and imaginary parts $16y - 9x = -25$ ----- (3) $16y + 8y - y = 9$ $16x + 7y = 9$ ----- (4) From (3) $x = \frac{16y + 25}{9}$ From (4) $16\left(\frac{16y + 25}{9}\right) + 7y = 9$ $256y + 400 + 63y = 81$ $319y = -319$ $y = -1$ $x = \frac{16(-1) + 25}{9} = 1$ $\therefore w = 1 - i$ $z = (4 + 2i)(1 + i) - 3i = 2 + 3i$	M1 – Either let $z = x + iy$, $x \in \mathbb{R}$, $y \in \mathbb{R}$ or $w = a + ib$ $a \in \mathbb{R}$, $b \in \mathbb{R}$ and attempt to solve the simultaneous equations M1 – Use of $w^* = x - iy$ and attempt to compare real and imaginary parts to form two simultaneous equation A1 – Either Obtain $w = 1 - i$ or $z = 2 + 3i$ A1 – All the answers correct
9(bi)	$\left \frac{u^2}{v^*} \right = \frac{ u ^2}{ v^* } = \frac{25}{6}$ $\arg\left(\frac{u^2}{v^*}\right) = 2\arg u - \arg v^* \text{ ----- (1)}$ $v = 6e^{i\frac{\pi}{2}} e^{-i\frac{\pi}{3}} = 6e^{i\frac{\pi}{6}}$ $\arg v^* = -\frac{\pi}{6}$ From (1)	B1 – Correct value of r i.e $\left \frac{u^2}{v^*} \right = \frac{25}{6}$ B1 – Correct $\arg v^* = -\frac{\pi}{6}$ M1 – Apply correct properties $\arg u^2 - \arg v^*$

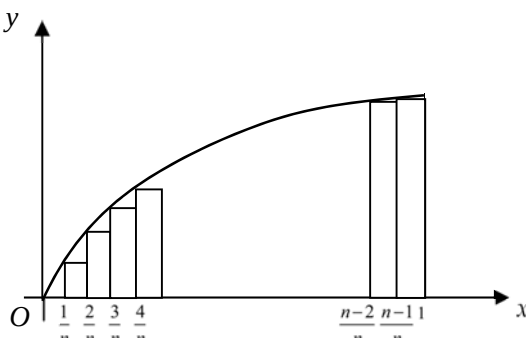
Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
	$\arg\left(\frac{u^2}{v^*}\right)$ $= 2\left(\frac{7\pi}{12}\right) - \left(-\frac{\pi}{6}\right)$ $= \frac{4\pi}{3}$ $\text{Arg}\left(\frac{u^2}{v^*}\right) = -2\pi + \frac{4\pi}{3} = -\frac{2\pi}{3}$ $\frac{u^2}{v^*} = \frac{25}{6} e^{-i\left(\frac{2\pi}{3}\right)}$	A1 – Correct answer for $\frac{u^2}{v^*}$
9 (bii)	$\frac{u^n}{v^*} = \frac{5^n e^{\frac{7}{12}n\pi i}}{6e^{-\frac{\pi}{6}i}}$ $= \frac{5^n}{6} e^{\left(\frac{7}{12}n\pi + \frac{\pi}{6}\right)i}$ To be purely imaginary $\frac{7n\pi + 2\pi}{12} = \frac{(2k+1)\pi}{2}, k \in \mathbb{Z}$ $12k + 4 = 7n$ $k = 2, n = 4$ $k = 9, n = 16$ $k = 16, n = 28$	M1 – Apply properties of $\arg\left(\frac{u^2}{v^*}\right)$ using previous part and equate it to either $\frac{(2k+1)\pi}{2}$ or $\frac{(2k-1)\pi}{2}$ OR to compare it with $\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2} \dots$ or to equate real part to zero and solve for n A1 – All three answers for n are correct by observation
10[14] (a)(i)	$\int x\sqrt{x+1} \, dx$ $= \frac{2x}{3}(x-1)^{\frac{3}{2}} - \int \frac{2}{3}(x-1)^{\frac{3}{2}} \, dx$ $= \frac{2x}{3}(x-1)^{\frac{3}{2}} + \frac{4}{15}(x-1)^{\frac{5}{2}} + c$ $u = x \Rightarrow u' = 1$ $v' = \sqrt{x-1} \Rightarrow v = \frac{2}{3}(x-1)^{\frac{3}{2}}$	M1 – Identify correct u and v' and use the correct formula for integration by parts A1 – Correct answer

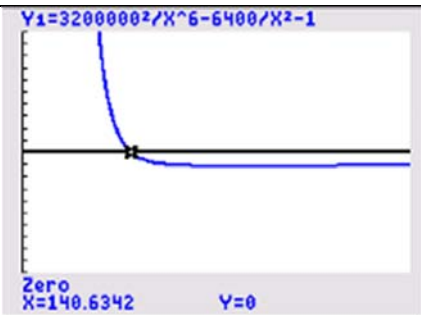
Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
10 (a)(ii)	 $y = \sqrt{a + x\sqrt{x-1}}$ $\sqrt{a+30} = \sqrt{a + x\sqrt{x-1}}$ $a+30 = a + x\sqrt{x-1}$ $900 = x^2(x-1)$ $x^3 - x^2 - 900 = 0$ $x = 10$ $V_c = \pi \int_1^{10} a + x\sqrt{x-1} \, dx$ $= \pi [ax]_1^{10} + \int_1^{10} x\sqrt{x-1} \, dx$ $= 9a\pi + \int_1^{10} x\sqrt{x-1} \, dx$ $= 9a\pi + \left[\frac{2}{3} x(x-1)^{\frac{3}{2}} \right]_1^{10} - \frac{2}{3} \int_1^{10} (x-1)^{\frac{3}{2}} \, dx$ $= 9a\pi + \pi \left[\frac{2}{3} x(x-1)^{\frac{3}{2}} - \frac{4}{15} (x-1)^{\frac{5}{2}} \right]_1^{10}$ $= 9a\pi + \pi \left[\frac{2700}{15} - \frac{972}{15} \right]$ $= \left(9a + \frac{1728}{15} \right) \pi \text{ units}^3$ $V = 9\pi (\sqrt{a+30})^2 - V_c$ $= 9\pi(a+30) - \left(9a + \frac{1728}{15} \right) \pi$ $= \frac{774\pi}{5} \text{ units}^3$	B1 – Attempt to solve for $x = 10$ M1 – Correct V_c or $9\pi(\sqrt{a+30})^2$ M1 – Find the required volume using $V = 9\pi(\sqrt{a+30})^2 - V_c$ A1 – Obtain Exact volume $\frac{774}{5}\pi$

Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
10 (c)(i)	 Area of 1st rectangle = $\frac{1}{n} \ln \left(1 + \frac{1}{n} \right)$ Area of 2nd rectangle = $\frac{1}{n} \ln \left(1 + \frac{2}{n} \right)$ Area of 3rd rectangle = $\frac{1}{n} \ln \left(1 + \frac{3}{n} \right)$ M Area of $(n-1)$th rectangle = $\frac{1}{n} \ln \left(1 + \frac{n-1}{n} \right)$ $A = \frac{1}{n} \left[\ln \left(1 + \frac{1}{n} \right) + \ln \left(1 + \frac{2}{n} \right) + \ln \left(1 + \frac{3}{n} \right) + \dots + \ln \left(1 + \frac{n-1}{n} \right) \right]$ $= \frac{1}{n} \ln \left[\left(1 + \frac{1}{n} \right) \left(1 + \frac{2}{n} \right) \left(1 + \frac{3}{n} \right) \dots \left(1 + \frac{n-1}{n} \right) \right]$ $= \frac{1}{n} \ln \left[\left(\frac{n+1}{n} \right) \left(\frac{n+2}{n} \right) \left(\frac{n+3}{n} \right) \dots \left(\frac{2n-1}{n} \right) \right]$ $= \frac{1}{n} \ln \left[\frac{(n+1)(n+2)(n+3) \dots (2n-1)}{n^{n-1}} \right]$	M1 – Attempt to find area of $n-1$ rectangles with the area of the first and the last rectangle correct AG1 – Apply laws of logarithms to show the expression given
10 (c)(ii)	$u = \ln(1+x) \Rightarrow u' = \frac{1}{1+x}$ $v' = 1 \Rightarrow v = x$ $\lim_{n \rightarrow \infty} A = \int_0^1 \ln(1+x) dx$	M1 – Use $\lim_{n \rightarrow \infty} A = \int_0^1 \ln(1+x) dx$ and apply integration by parts correctly

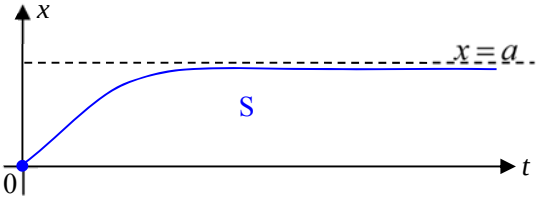
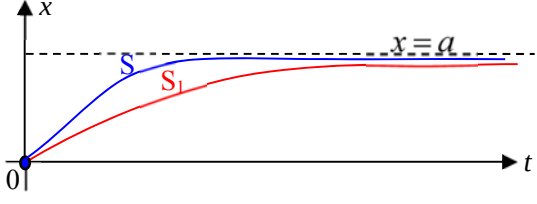
Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
	$= [x \ln(1+x)]_0^1 - \int_0^1 1 - \frac{1}{1+x} dx$ $= [x \ln(1+x)]_0^1 - [x - \ln(1+x)]_0^1$ $= \ln 2 - [1 - \ln 2]$ $= 2 \ln 2 - 1 \text{ units}^2$	M1 – Rewrite as $\int_0^1 1 - \frac{1}{1+x} dx$ and integrate correctly A1 – Correct answer in exact form
11(i)	Floor area $xy = 4000 \Rightarrow y = \frac{4000}{x}$ Area $A = 2(4x) + 2x \sqrt{(0.01y^2)^2 + \left(\frac{y}{2}\right)^2}$ $= 8x + 2x \sqrt{\left(0.01 \frac{4000^2}{x^2}\right)^2 + \left(\frac{4000}{2x}\right)^2}$ $= 8x + 2x \sqrt{\frac{160000^2}{x^4} + \frac{2000^2}{x^2}}$ $= 8x + 4000 \sqrt{\frac{6400}{x^2} + 1}$	M1 – Express A in terms of x and y M1 – Substitute $y = \frac{4000}{x}$ AG1 – Able to simplify to given expression
11(ii)	$\frac{dA}{dx} = 8 + 2000 \left(\frac{6400}{x^2} + 1 \right)^{-1/2} \left(-\frac{12800}{x^3} \right) = 0$ $\left(\frac{6400}{x^2} + 1 \right)^{-1/2} \frac{25600000}{x^3} = 8$ $\frac{3200000}{x^3} = \sqrt{\frac{6400}{x^2} + 1}$ $\frac{1.024 \times 10^{13}}{x^6} = \frac{6400}{x^2} + 1$ $a = 1.024 \times 10^{13}$	M1 – Apply Chain Rule to get $\frac{1}{2}(f(x))^{-1/2} \left(-\frac{k}{x^3} \right)$. M1 – Equate $\frac{dA}{dx} = 0$ and simplify to given expression A1 – Correct value of a in any form e.g. $a = 3200000^2 = 20^{10}$.
11(iii)	 By GC, $x = 140.6342 = 141$ to 3 s.f.	B1 – Correct value of x to 3 s.f.

Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme												
	Method 1: <table><tr><td>x</td><td>140.63⁻</td><td>140.63</td><td>140.63⁺</td></tr><tr><td>$\frac{dA}{dx}$</td><td>-ve</td><td>0</td><td>+ve</td></tr><tr><td></td><td>\</td><td>-</td><td>/</td></tr></table> Method 2: $\frac{d^2 A}{dx^2} = \left(\frac{6400}{x^2} + 1 \right)^{-1/2} \left(\frac{76800000}{x^4} \right)$ $+ \left(\frac{6400}{x^2} + 1 \right)^{-3/2} \left(-\frac{12800000}{x^3} \right) \left(-\frac{12800}{x^3} \right)$ > 0 when $x = 140.6342$ Hence A is minimum when $x = 141$ to 3 s.f. Minimum Cost = $\left[8(140.6342) + 4000 \sqrt{\frac{6400}{140.6342^2} + 1} \right] 2.50$ = \$14317.43 (nearest cents) = \$14300 (to 3 s.f.)	x	140.63 ⁻	140.63	140.63 ⁺	$\frac{dA}{dx}$	-ve	0	+ve		\	-	/	B1 – Apply 1st or 2nd derivative test [only award mark if value of x is correct] B1 – Correct minimum cost
x	140.63 ⁻	140.63	140.63 ⁺											
$\frac{dA}{dx}$	-ve	0	+ve											
	\	-	/											
11(iv)	From (ii), $\frac{dA}{dx} = 8 - \left(\frac{6400}{x^2} + 1 \right)^{-1/2} \frac{25600000}{x^3}$ When $x = 200$, $\frac{dA}{dt} = \frac{dA}{dx} \frac{dx}{dt}$ $= \left[8 - \left(\frac{6400}{200^2} + 1 \right)^{-1/2} \frac{25600000}{200^3} \right] 2$ = 10.0577 = 10.1 unit²/min (3 s.f.)	M1 – Attempt Chain rule to find $\frac{dA}{dx}$ or apply implicit differentiation to obtain an equation involving $\frac{dA}{dt}$ and $\frac{dx}{dt}$ M1 – Attempt to substitute $x = 200$ and $\frac{dx}{dt} = 2$ A1 – Correct answer to 3 s.f.												
12(i)	the concentration of A and B at time t are $(a - x)$ and $(b - x)$ mol/dm ³ respectively	B1 – both answers correct												
12(ii)	$\frac{dx}{dt} = k(a - x)(b - x)$, $k \in \mathbb{R}^+$	B1 – Correct answer												
12(iii)	Max value for x is a , $Q \frac{dx}{dt} = 0$ and after $x = a$ there is no more concentration of substance A for reaction to continue.	B1 – Correct Max value for x is a B1 – state correct reason												

Marking Scheme for HCI 2018 Prelim Paper 1

Qn	Suggested Solution	Marking Scheme
12(iv)	$\frac{dx}{dt} = k(a-x)^2$ $\int (a-x)^{-2} dx = kt$ $(a-x)^{-1} = kt + C$ $x = a - \frac{1}{kt + C} \text{ --- (1)}$ When $t = 0, x = 0 \Rightarrow C = \frac{1}{a}$ $x = a - \frac{1}{kt + \frac{1}{a}}$ $x = a - \frac{a}{akt + 1}$ 	M1 – correct method of integration $\int (a-x)^{-2} dx$ A1 – correct expression (1) accept Answer without constant M1 – Use of initial value to find constant A1 – Correct particular solution $x = a - \frac{a}{akt + 1}$ A1 – correct graph with asymptote and initial value labelled
12(v)	$\frac{dx_1}{dt} = 4 \frac{dx}{dt}$	B1
12(vi)		B2 – one mark for each curve showing clearly relative position Do not Accept graphs with S and S₁ overlapping each other at the tail ends
12(vii)	From the graph, $\alpha < \beta$. Q $\frac{dx_1}{dt}$ for S₁ > $\frac{dx}{dt}$ for S and both curves start from the origin.	B1 – correct $\alpha < \beta$ B1 – correct reason explain using the concentration of the